THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

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CLASS 10 — MATHS · EASY LEVEL CHAPTER TEST SET 1

Chapter 14: **Probability** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Easy level: direct, single-step questions.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. A die is thrown once. The probability of getting an even number is: 1

(a) $\frac{1}{3}$ **(b)** $\frac{3}{7}$ **(c)** $\frac{2}{3}$ **(d)** $\frac{1}{2}$

2. A die is thrown once. The probability of getting a multiple of 3 is: 1

(a) $\frac{1}{2}$ **(b)** $\frac{1}{6}$ **(c)** $\frac{2}{3}$ **(d)** $\frac{1}{3}$

3. A coin is tossed two times. The probability of getting two heads is: 1

(a) 0 **(b)** $\frac{3}{4}$ **(c)** $\frac{1}{2}$ **(d)** $\frac{1}{4}$

4. A bag contains 3 red, 5 blue balls. One ball is drawn at random. The probability that it is red is: 1

(a) $\frac{1}{2}$ **(b)** $\frac{3}{8}$ **(c)** $\frac{1}{4}$ **(d)** $\frac{5}{8}$

5. 10 cards numbered 1 to 10 are placed in a box, mixed well and one is drawn at random. The probability that the number drawn is a multiple of 3 is: 1

(a) $\frac{1}{5}$ **(b)** $\frac{2}{5}$ **(c)** $\frac{3}{10}$ **(d)** $\frac{7}{10}$

6. The probability of an event E is $\frac{1}{4}$. The probability of the event 'not E' is: 1

(a) $\frac{5}{4}$ **(b)** $\frac{1}{8}$ **(c)** $\frac{3}{4}$ **(d)** $\frac{1}{4}$

7. The probability of a sure event is: 1

(a) 2 **(b)** 1 **(c)** $\frac{1}{2}$ **(d)** 0

8. A die is thrown once. The probability of getting an odd number is: 1

(a) $\frac{3}{7}$ **(b)** $\frac{1}{2}$ **(c)** $\frac{2}{3}$ **(d)** $\frac{1}{3}$

9. A die is thrown once. The probability of getting a perfect square is: 1

(a) $\frac{1}{6}$ **(b)** $\frac{2}{3}$ **(c)** $\frac{1}{3}$ **(d)** $\frac{1}{2}$

10. A coin is tossed two times. The probability of getting at least one head is: 1

(a) $\frac{1}{2}$ **(b)** 1 **(c)** $\frac{3}{4}$ **(d)** $\frac{1}{4}$

SECTION — B (Very Short Answer Type) 4 × 2 = 8

Answer in about 20–30 words each.

11. A die is thrown once. Find the probability of getting a number greater than 6. 2

12. A coin is tossed two times. Find the probability of getting two heads. 2

13. A bag contains 5 red, 3 blue balls. One ball is drawn at random. Find the probability that it is blue. 2

14. A die is thrown once. Find the probability of getting a number not less than 5. 2

SECTION — C (Short Answer Type) 4 × 3 = 12

Answer in about 40–60 words each.

15. In a lottery there are 10 prizes and 25 blanks. A ticket is drawn at random. Find the probability of getting a prize. 3

16. A lot of 144 ball pens contains 20 defective pens. One pen is drawn at random from the lot. Find the probability that the pen is good (not defective). 3

17. A class has 40 students, of whom 24 are girls. A student is selected at random to be the monitor. Find the probability that a boy is selected. 3

OR

In a lottery there are 8 prizes and 32 blanks. A ticket is drawn at random. Find the probability of getting a prize. 3

18. A lot of 100 ball pens contains 12 defective pens. One pen is drawn at random from the lot. Find the probability that the pen is good (not defective). 3

OR

A class has 30 students, of whom 12 are girls. A student is selected at random to be the monitor. Find the probability that a boy is selected. 3

SECTION — D (Long Answer Type) 2 × 5 = 10

Answer in about 100–150 words each.

19. A bag contains 3 red, 4 blue, 3 green balls. One ball is drawn at random. Find the probability that the ball is: 5
(i) red,
(ii) not blue,
(iii) yellow (in this order).

OR

A box contains 12 cards numbered 1 to 12. One card is drawn at random. Find the probability that the number is: 5
(i) a number less than 10,
(ii) a prime number,
(iii) a number greater than 12 (in this order).

20. A bag contains 5 red, 7 blue, 8 green balls. One ball is drawn at random. Find the probability that the ball is: 5
(i) green,
(ii) red or blue,
(iii) white (in this order).

OR

A box contains 20 cards numbered 1 to 20. One card is drawn at random. Find the probability that the number is: 5
(i) a multiple of 3,
(ii) a prime number,
(iii) a multiple of 4 (in this order).

END OF QUESTION PAPER Total: 40 Marks